西北工业大学研究生学位论文选题报告表

Northwestern Polytechnical University

Graduate Degree Dissertation Title Selection Report Form

学　号 **Student No.**	_____
学　院 **School**	计算机学院 School of Computer Science
学科、专业 **Discipline / Specialty**	计算机科学与技术 Computer Science and Technology
姓　名 **Name**	SAPARGALI ALMAZ
学位级别 **Degree**	硕士 Master
导　师 **Supervisor**	_____
培养类别 **Category**	留学研究生 Overseas Postgraduates
报告日期 **Date of Report**	2026 年 9 月 / September 2026

论文题目 **Dissertation Title**	Surrogate Model Selection for Reinforcement-Learning Control of Building HVAC Systems: A Step-Size Criterion and a Role-Separated Hybrid Architecture

论文类型 **Dissertation Type** **(在有关项目下作记号 √)**	基础研究 Fundamental Research	应用研究 Application Research	工程技术 Engineering Technology	跨学科研究 Research Cross Discipline
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1. Research Background

Buildings account for roughly one third of global final energy use, and heating, ventilation and air conditioning (HVAC) equipment is the largest single consumer within them. Improving how this equipment is controlled is therefore one of the few interventions that can reduce energy demand at scale without replacing physical infrastructure. Reinforcement learning (RL) is attractive for this task because it learns a control policy directly from interaction, without requiring an explicit analytical model of the building, and can in principle balance thermal comfort against energy cost in a way that fixed rule-based logic cannot.

A practical obstacle stands in the way. Training an RL controller requires millions of environment interactions. A high-fidelity building emulator written in Modelica advances at approximately 21 simulation steps per second on ordinary hardware, which places direct training far beyond any practical time budget. The response adopted across essentially the whole HVAC RL literature is to fit a fast learned surrogate of the building, train the policy against that surrogate, and only then evaluate the resulting controller on the emulator or the real building. The surrogate plays the role of a digital twin used as a training environment.

This makes one design decision unavoidable: among several candidate surrogates, which one should be used for training? In current practice the answer is decided by predictive error. Papers on building surrogates report multi-step rollout error, error distributions and calibration statistics, and the model with the lowest error is carried forward. The heuristic is intuitive and, to the best of our knowledge, universal in the building-energy literature. It is also untested. Predictive error is measured on held-out trajectories generated by some other controller, whereas a training environment is used as a closed-loop transition function on states that the learning policy itself induces. Nothing guarantees that the two orderings agree.

Preliminary work carried out for this dissertation shows that they do not. On the open BOPTEST benchmark, a physically calibrated grey-box twin predicts far better than a compact black-box network — a 24-hour rollout error of 0.644 °C against 1.557 °C — yet the policy trained on the accurate twin leaves the comfort band for more than 77 % of the evaluation horizon on every random seed, while the policy trained on the less accurate network is the only single-model controller in the study that is usable. Retraining the same black-box architecture on a finer-resolution corpus makes it strictly more accurate (0.876 °C) and equally unusable, which rules out model class as the explanation.

The research problem addressed by this dissertation follows directly. If predictive accuracy does not predict whether a surrogate will train a usable controller, then what property does? The question matters beyond a single case study: any project that trains a controller against a learned model of a physical system faces the same selection decision, and currently makes it on a criterion that has not been validated for that purpose.

1.1 Proposed Research Content of the Paper

The dissertation develops and validates a selection criterion for surrogate training environments, together with an architecture that acts on that criterion. The work is organised into three connected parts.

The first part characterises the surrogates themselves. Two families are constructed for the same building: a compact black-box neural predictor of the temperature and power increments, and a physics-informed grey-box model built on a lumped-capacity resistance–capacitance (RC) balance whose unmodelled heat flow is supplied by a neural residual term and whose parameters are identified from telemetry by a staged inverse procedure. Predictive fidelity is measured over multiple horizons, and the corpus resolution is varied so that model class and temporal resolution can be separated as explanatory factors.

The second part measures control utility rather than predictive accuracy. The same controller architecture, reward function, training budget and evaluation protocol are applied to every surrogate, so that the only variable is the training environment. Controllers are then evaluated zero-shot on the live emulator

over two fourteen-day windows, a peak-heating window and a typical window, using a maintenance score that combines time spent outside the comfort band with the severity of the excursions. This part also identifies the property that does predict utility — the magnitude of the surrogate's per-step state increment, expressed as a scale-free roughness of the action-response surface — and isolates it causally with a rescaling control that changes the integration step while holding that roughness fixed.

The third part concerns generalisation. Two boundary tests are performed. The first asks whether the effect is specific to policy-gradient learning by replacing the learned controller with a receding-horizon model predictive control (MPC) planner that solves an optimisation at every step. The second asks whether the methodology transfers to buildings other than the one it was developed on, using three independent hydronic test cases that differ in heat source, heat-transfer medium and scale.

Alongside these, the dissertation develops the control layer itself in a multi-objective formulation, so that the trade-off between occupant comfort and energy use is exposed as an operator-adjustable preference at run time rather than fixed before training, and a role-separated hybrid architecture that resolves the conflict between smoothness and fidelity by assigning the two requirements to different models instead of compromising within one.

1.2 Feasibility of the Research Plan and Implementation of Research Conditions

The plan is feasible on three grounds: the experimental platform is already in place, the computational requirements are modest and measured rather than estimated, and a substantial portion of the work has already been completed and externally submitted.

All experiments run on BOPTEST, the open building-emulator benchmark developed under IBPSA Project 1. This choice is deliberate: BOPTEST supplies Modelica emulators, a built-in reference controller and a fixed set of key performance indicators, so results can be reproduced and contested by other groups rather than resting on a proprietary building model. The benchmark is deployed locally as a containerised service, and the training code runs in its own container on the same network, an arrangement that is already operational.

Computational requirements have been measured, not assumed. The emulator advances at 21 steps per second, whereas the black-box surrogate reaches 4626 and the hybrid environment 1787, giving speed-ups of 220 and 85 times respectively. These figures were obtained on a six-core laptop-class processor with 32 GiB of memory and no graphics accelerator. A full training cycle that would require approximately 66 hours of emulator time completes in well under an hour on the surrogate. No specialised hardware is therefore required, which removes the most common source of schedule risk in deep-learning projects.

The software stack consists of Python 3.11, PyTorch and Stable-Baselines3, all mature and openly available. The full experimental codebase, the trained surrogate checkpoints and the derived evidence artifacts already exist in a version-controlled repository, together with a documented reproduction procedure at three levels: inspection of committed artifacts, regeneration of every figure and table without the emulator, and full re-execution of the experiments. Every numerical value reported in this proposal is read from that evidence rather than restated from memory.

Finally, the risk that the topic proves unproductive is low, because the central result has already been obtained, replicated across random seeds and written up. A manuscript covering the core of the work has been submitted to IEEE Access (manuscript ID Access-2026-42535). The remaining work is extension and consolidation rather than open-ended exploration.

2. Research Status

2.1 Physics-Based and Grey-Box Building Models

Reduced-order thermal models of buildings have a long history in control engineering. Resistance–capacitance networks represent the building envelope as a small set of thermal resistances and capacitances whose parameters are identified from measured data. Their appeal is interpretability and data efficiency: the identified parameters correspond to physical quantities, and the structure constrains extrapolation. Recent work extends these models with learned residual terms, neural ordinary differential equations and physics-informed neural networks, combining physical structure with the flexibility of learned components. This line of work is evaluated almost exclusively on predictive accuracy, which is the assumption the present research examines.

2.2 Learned Surrogates and Deep Learning for Building Dynamics

Purely learned surrogates trade interpretability for speed and modelling capacity. Protocols have been proposed for developing and evaluating neural-network surrogates of building simulations, covering sample generation, feature selection, architecture choice and validation. Hardware-accelerated implementations push inference latency low enough for real-time use. Across this literature, the evaluation criterion remains predictive: the surrogate is judged by how closely it reproduces held-out simulation output. The possibility that a surrogate might be accurate and simultaneously unsuitable as a training environment is not addressed.

2.3 Reinforcement Learning for HVAC Control

Reinforcement learning for building control has grown rapidly, with surveys documenting applications to setpoint scheduling, supply-temperature control and multi-zone coordination. Reported work spans single-agent and multi-agent formulations, hierarchical decomposition across timescales, safety-constrained learning, and the use of forecast information in the observation vector. Multi-objective formulations condition the policy on a preference vector so that a single trained policy can serve a range of comfort–energy trade-offs. Almost all of this work trains against a simulator or a learned surrogate; the choice of that training environment is typically reported as an implementation detail rather than investigated as a design variable.

2.4 Limitations of Current Research

Four gaps motivate the present work. First, the criterion used to select a training surrogate has never been validated against the outcome it is meant to predict, namely the quality of the resulting controller. Second, the failure modes of a surrogate used in closed loop are poorly characterised: predictive metrics are computed on off-policy trajectories and are blind to defects that a controller will actively seek out, including a model whose response to a control command has the wrong sign. Third, transfer studies usually report whether a method works on a new building without separating which component transferred — the model, the calibration procedure or the policy — so the boundary of applicability remains unclear. Fourth, results are frequently reported for a single random seed, which for policy-gradient methods is insufficient to distinguish an effect from initialisation noise.

3. Methodology and Roadmap

The methodology is organised as three experimental blocks with a common evaluation protocol, followed by two boundary tests. A discipline of pre-registration runs through all of them: each hypothesis, its pass criterion and the actuator adapters are written down and committed to version control before the corresponding runs are executed, and the commit identifier serves as an audit anchor. This prevents the criterion from being adjusted after the outcome is known, which is the principal threat to validity in a study whose central claim is negative.

Block 1 establishes predictive fidelity. Both surrogate families are trained on telemetry collected from the emulator, and their multi-horizon rollout error, residual distributions and physical consistency are

measured. The grey-box model is calibrated by a three-stage inverse procedure that aligns telemetry, identifies the zone capacitance and fits the residual heads. The corpus resolution is treated as an explicit experimental factor so that the effect of temporal resolution can be separated from the effect of model class.

Block 2 measures control utility. A thermostatic proximal-policy-optimisation controller is trained on each surrogate under identical conditions and evaluated zero-shot on the live emulator over the two seasonal windows. The maintenance score combines the fraction of time outside the comfort band with the severity of excursions; a value above one marks a controller that cannot be placed in service. This block also contains the mechanism experiment: the action-response surface of each surrogate is swept, its curvature-to-slope ratio computed, and a rescaling control applied that alters the integration step while preserving that ratio, so that step size and surface roughness are decoupled.

Block 3 addresses transfer. The complete pipeline is applied to three hydronic test cases that were not used during development. Transfer is decomposed into two independent mechanisms and reported separately: whether the calibration procedure transfers, measured by the reduction in rollout error after recalibration, and whether a frozen controller transfers without retraining, measured against a case-specific reference controller.

Two boundary tests complete the design. The first replaces the learned controller with a receding-horizon MPC planner, which never performs gradient ascent on policy parameters, in order to test whether the effect is specific to policy-gradient learning. The second is a directional-validity audit that samples states and asks, for each actuated input separately, whether increasing the command moves the zone temperature in the physically correct direction.

3.2 Technical Implementation Strategy

The experimental platform is a containerised BOPTEST service exposing the emulator through an HTTP interface, with the training code in a separate container on the same network. Every environment backend — the emulator itself and each surrogate — is placed behind one common interface, so that switching the training environment changes a single configuration entry and nothing else in the training code. This is what makes the central comparison clean: the controller, reward, budget and evaluation are provably identical across backends.

The black-box surrogate is a compact dual-head network predicting the zone-temperature increment and the electrical power draw. The grey-box surrogate combines a lumped-capacity balance with a neural residual head and is calibrated in three stages against telemetry. The hybrid environment couples them by role: the black-box model supplies the rollout dynamics, and the frozen grey-box model is evaluated forward-only on the same state–action pair and contributes a penalty proportional to the disagreement between the two predictions. The grey-box model never participates in the rollout and never enters the policy loss, which is what distinguishes this architecture from model ensembling.

Controllers are trained with proximal policy optimisation. The multi-objective variant conditions the policy on a preference vector supplied as part of the observation, so a single trained policy covers the whole comfort–energy trade-off and the operating point can be moved at run time without retraining. All central comparisons are replicated over at least three random seeds, and the censor-weight sweep is replicated at every swept value, so that reported differences can be compared against seed dispersion rather than asserted.

The roadmap proceeds in three phases. The first, already largely complete, covers surrogate construction, the fidelity–utility comparison and the mechanism experiment. The second covers the boundary tests, the multi-objective control layer and the seed replication programme. The third covers consolidation: the remaining experiments identified below, the dissertation text, and the defence.

4. Innovations and Challenges

4.1 Key Innovations

The first contribution is a negative result with a constructive replacement. The dissertation demonstrates that predictive error, the criterion universally used to select a surrogate training environment, is not merely uninformative for that purpose but inverted over part of the range, and it identifies a criterion that does work. The operative property is the magnitude of the surrogate's per-step state increment, expressed as the scale-free roughness of its action-response surface. The measured value for the usable surrogate is 0.169, against 9.4 and 7.9 times that value for the two surrogates that fail. Crucially, this quantity is measured on the surrogate alone, before any controller is trained, which makes it usable as a selection criterion rather than a post-hoc explanation.

The second contribution is the role-separated hybrid architecture. Existing practice treats smoothness and fidelity as a trade-off to be resolved within a single model. The proposed architecture assigns them to different models: the black-box surrogate supplies smooth rollout dynamics, while the frozen physics-informed twin acts solely as a forward-only model-disagreement reward censor. This yields the best worst-window result in the study, a maintenance score of 0.041 on the typical window against 1.102 for the accurate twin used directly, while retaining a speed-up of 85 times over the live emulator.

The third contribution is a diagnostic that predictive metrics cannot provide. The directional-validity audit checks, for each actuated input separately, whether the surrogate's response to a control command has the correct sign. Applied to the finer-resolution surrogate it reveals that the model is inverted in roughly nine sampled states out of ten — a hotter supply-temperature command predicting a colder zone — reproducibly across four independent training draws, while its rollout error improves in every one of them. A monotonicity constraint added to training restores the sign on the constrained channel at a still lower rollout error but relocates the defect to the fan channel, which establishes that such an audit must be performed per input and before controller training.

The fourth contribution is methodological. Hypotheses and their pass criteria are pre-registered with version-control anchors before execution, and outcomes are reported as they occur. Of five pre-specified hypotheses, three are falsified, including two predictions made by the authors themselves. One planned experiment was executed and then invalidated by a defect found in our own planner; that run is retained in the record rather than discarded, because deleting it and quietly re-running would leave the pre-registration looking untouched when a completed run had already been observed. This standard of evidence is uncommon in the applied building-control literature and is itself a contribution of the dissertation.

4.2 Technical Challenges

The principal difficulty is that the central claim is negative, and negative claims are easy to produce by accident. A result showing that an accurate model trains a poor controller could equally be explained by a bug in the training pipeline, an unlucky seed, or an unfair comparison. The design answers this with three countermeasures: a single common interface behind which every backend sits, so that nothing but the environment differs; replication across random seeds with dispersion reported alongside every headline number; and pre-registered pass criteria. The negative control is part of the design rather than an afterthought — a planner running on a knowingly sign-inverted model is required to fail, and does.

A second challenge is disentangling confounded factors. Temporal resolution, model class and response-surface roughness co-vary in the natural experiment, so observing that a finer-resolution model fails does not by itself identify the cause. The rescaling control is the instrument that separates them, and constructing it correctly — changing the integration step while provably preserving scale-free roughness — is the most delicate part of the experimental design.

A third challenge is the fragility of the evaluation itself. During the work an MPC run produced identical results across three different surrogates, which is impossible unless the planner is ignoring them; the cause was a saturation defect in our own planner rather than a property of the models. Detecting such

failures requires sanity checks that are independent of the quantity being measured, and building them into the harness is an ongoing requirement rather than a one-off task.

A fourth challenge concerns the multi-objective layer. The preference-conditioned controller is functional but not yet reproducible to the standard the dissertation sets elsewhere: extending replication to five seeds gives a maintenance score of 0.187 ± 0.078 at the neutral preference point, which fails a pre-specified stability test. Establishing whether this reflects an inherent difficulty of preference-conditioned training or a correctable defect in the training procedure is a specific open task for the remaining period.

5. Current Progress

Substantial preliminary results have already been obtained, and they are the basis on which this topic is proposed.

The experimental platform is complete and operational: a containerised BOPTEST deployment, both surrogate families implemented and calibrated, the hybrid environment, and a common backend interface behind which they all sit. Performance has been benchmarked and is reported above.

The central comparison is complete. Four training environments were compared under identical conditions, with the results shown below. All values are live closed-loop measurements on the emulator, averaged over the two seasonal windows, with the controllers replicated over three random seeds.

Table 1 — Predictive fidelity against control utility.

Training environment	24 h rollout error, °C	Score, peak window	Score, typical window	Verdict
Black-box surrogate, 1 h step	1.557	0.072	0.095	usable
Black-box surrogate, 15 min step	0.876	1.142	1.211	not usable
Grey-box twin, calibrated	0.644	1.046	1.102	not usable
Role-separated hybrid	1.557	0.087	0.041	usable

The mechanism has been measured rather than hypothesised. The relative roughness of the action-response surface is 0.169 for the usable surrogate and 1.578 and 1.338 for the two that fail, and the rescaling control reproduces the collapse while holding roughness fixed, establishing the relationship as causal rather than correlational.

Both boundary tests have been executed. The MPC planner reproduces the same inverted ordering as the learned controllers, which falsifies our own pre-registered prediction that the effect would be specific to policy-gradient search and widens the scope of the finding to optimisation through a surrogate in general. On the transfer side, recalibration reduces rollout error by 60.2, 87.4, 87.8 % on the three hydronic cases, and the re-identified zone capacitance is consistently 1.89 to 1.95 times the baseline value, which is physically consistent with the greater thermal inertia of a water circuit. Frozen-policy transfer, by contrast, succeeds on only one of the three cases. Transfer therefore resolves into a component-level boundary: the calibration pipeline transfers, the policy does not.

The multi-objective control layer is implemented and evaluated. Widening the observation interface from five instantaneous features to seventeen forecast-augmented features moves the controller from unusable to usable — rollout error falls from 4.96 to 0.72 °C, comfort violation from 74.5 % to 4.9 %, and the maintenance score from 1.046 to 0.099. This identifies observability, not reward shaping, as the binding constraint in that formulation.

Finally, a manuscript covering the core of this work has been submitted to IEEE Access (Access-2026-42535), and earlier results were presented as a conference talk at the WCCM–ECCOMAS Congress 2026

in Munich. A reproduction package containing the code, the trained checkpoints and the derived evidence has been assembled and verified from a clean checkout.

6. Dissertation Plan

October 2026 – December 2026: Chapters 1 and 2 (Problem formulation, surrogate construction and predictive fidelity). Consolidate the literature review with emphasis on the selection criterion used in prior surrogate work. Write up the construction and calibration of both surrogate families, the staged inverse identification procedure, and the multi-horizon fidelity evaluation. Complete the sensitivity analysis of the calibration stages that is currently only partially documented. Milestone: Chapters 1 and 2 drafted; the fidelity evaluation reproducible end-to-end from committed artifacts.

January 2027 – February 2027: Chapter 3 (Control utility and mechanism). Write up the central fidelity –utility comparison, the mechanism experiment and the rescaling control. Extend the directional-validity audit to the grey-box family, which has so far been audited only as an admission gate for the planner. Complete the derivative-free planner experiment identified as the natural follow-up to the MPC result: the current planner is itself a first-order method through a differentiable surrogate, so it separates gradient-based optimisation from other synthesis routes rather than separating planning from learning. Milestone: Chapter 3 drafted; the derivative-free planner result obtained and reported whichever way it comes out.

March 2027 – April 2027: Chapter 4 (Architecture, multi-objective control and transfer). Write up the role-separated hybrid architecture and the censor-weight sensitivity study. Address the open reproducibility issue in the multi-objective layer by extending seed replication and investigating whether the dispersion at the neutral preference point is inherent or correctable. Write up the three-case transfer study and the component-level boundary. Milestone: Chapter 4 drafted; the multi-objective stability question resolved and reported.

May 2027 – June 2027: Chapter 5 and defence. Integrate the chapters, write the discussion of threats to validity and the limitations statement, and complete the conclusions. Prepare the defence presentation and conduct practice sessions. Submit the final dissertation and defend. Milestone: complete dissertation submitted and successfully defended by the end of June 2027.

Two publication milestones run alongside the writing. The IEEE Access manuscript is currently under review and any revision required by the reviewers will be handled within the first phase. A second manuscript covering the transfer study and the directional-validity diagnostic is planned for submission during the third phase.

7. References

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